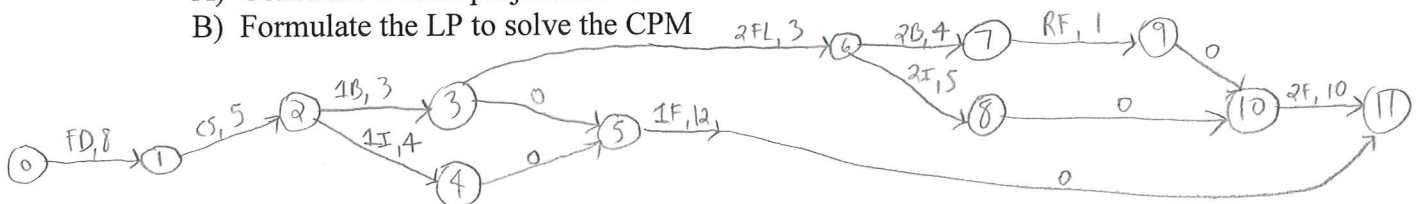


Construction of a small two-story house involves the tasks listed in the following table. The table also shows the estimated duration of each task in days and the tasks that must complete before it can begin.

Activity	Duration (in days)	Immediate Predecessors
FD. Foundation	8	-
CS. Concrete Slab	5	FD
1B. First Bearing Walls	3	CS
1I. First internal walls	4	CS
1F. First finishing	12	1B, 1I, FL
2FL. Second floor	3	1B
2B. Second Bearing Walls	4	2FL
2I. Second internal walls	5	2FL
2F. Second finishing	10	2B, 2I, RF
RF. Roof	1	2B

To effectively administer the product development, the manager of the project would like to know the following:

- Construct a CPM project network
- Formulate the LP to solve the CPM



$$\begin{aligned}
 t_1 &\geq t_0 + 8 \\
 t_2 &\geq t_1 + 5 \\
 t_3 &\geq t_2 + 3 \\
 t_4 &\geq t_2 + 4 \\
 t_5 &\geq t_3 \\
 t_5 &\geq t_4 \\
 t_6 &\geq t_3 + 3 \\
 t_7 &\geq t_6 + 4 \\
 t_8 &\geq t_6 + 5 \\
 t_9 &\geq t_7 + 1 \\
 t_{10} &\geq t_8 \\
 t_{10} &\geq t_9 \\
 t_{11} &\geq t_{10} + 10 \\
 t_{11} &\geq t_5 + 12 \\
 \min \quad t_{11}
 \end{aligned}$$